

Biological evaluation of medical devices

Part 4: Selection of tests for interactions with blood

INTRODUCTION

The selection and design of test methods for the interactions of medical devices with blood should take into consideration device design, materials, clinical utility, usage environment and risk benefit. This level of specificity can only be covered in vertical standards.

The initial source for developing this part of ISO 10993 was the publication, *Guidelines for blood/material interactions*, Report of the National Heart, Lung, and Blood Institute [29]; chapters 9 and 10. This publication has since been revised [32].

Biological evaluation of medical devices

Part 4: Selection of tests for interactions with blood

1 Scope

This part of ISO 10993 provides general requirements for evaluating the interactions of medical devices with blood.

It describes

- a) a classification of medical and dental devices that are intended for use in contact with blood, based on the intended use and duration of contact as defined in ISO 10993-1,
- b) the fundamental principles governing the evaluation of the interaction of devices with blood,
- c) the rationale for structured selection of tests according to specific categories, together with the principles and scientific basis of these tests.

Detailed requirements for testing cannot be specified because of limitations in the knowledge and precision of tests for interactions of devices with blood. This part of ISO 10993 describes biological evaluation in general terms and may not necessarily provide sufficient guidance for test methods for a specific device.

2 Normative references

The following normative documents contain provisions which, through reference in this text, constitute provisions of this part of ISO 10993. For dated references, subsequent amendments to, or revisions of, any of these publications do not apply. However, parties to agreements based on this part of ISO 10993 are encouraged to investigate the possibility of applying the most recent editions of the normative documents indicated below. For undated references, the latest edition of the normative document referred to applies. Members of ISO and IEC maintain registers of currently valid International Standards.

ISO 10993-1:1997, *Biological evaluation of medical devices — Part 1: Evaluation and testing*

ISO 10993-2:1992, *Biological evaluation of medical devices — Part 2: Animal welfare requirements*

3 Terms and definitions

For the purposes of this part of ISO 10993, the terms and definitions given in ISO 10993-1 and the following apply.

3.1

blood/device interaction

any interaction between blood or any component of blood and a device resulting in effects on the blood, or on any organ or tissue, or on the device

NOTE Such effects may or may not have clinically significant or undesirable consequences. Annex A contains further information on these interactions.

3.2

ex vivo

term applied to a test system that shunts blood directly from a human subject or test animal into a test chamber located outside the body

NOTE If using an animal model, the blood may be shunted directly back into the animal (recirculating) or collected into test tubes for evaluation (single pass).

3.3

thrombosis

in vivo phenomenon resulting in the partial or complete occlusion of a vessel or device by a thrombus

NOTE 1 Characterization of thrombosis includes *ex vivo* and *in vivo* methods, in either animals or the clinical setting.

NOTE 2 A thrombus is composed of a mixture of red cells, aggregated platelets, fibrin and other cellular elements.

4 Abbreviated terms

Bb product of alternative pathway complement activation

β-TG beta-thromboglobulin

C4d product of classical pathway complement activation

C3a, C5a (active) complement split products from C3 and C5

CD62L L-selectin

CH-50 50% total haemolytic complement

CT computerized tomography

D-Dimer specific fibrin degradation products (F XIII cross-linked fibrin)

ECMO extracorporeal membrane oxygenator

ELISA enzyme/linked immunosorbent assay

EM electron microscopy

FDP fibrin/fibrinogen degradation products

FPA fibrinopeptide A

F1+2 prothrombin activation fragment 1 + 2

iC3b product of central C complement activation

IVC inferior vena cava

MRI magnetic resonance imaging

PAC-1 monoclonal antibody which recognizes the activated form of platelet surface glycoprotein IIb/IIIa

PET positron emission tomography

PF-4 platelet factor 4

PRP platelet-rich plasma

PT prothrombin time

PTT partial thromboplastin time

P-selectin receptor exposed during either platelet or endothelial cell release reaction

RIA radioimmunoassay

S-12 monoclonal antibody, which recognizes the alpha-granule membrane component

P-selectin exposed

during the platelet release reaction

SC5b-9 product of terminal pathway complement activation

TAT thrombin-antithrombin complex

TCC terminal complement complex

TT thrombin time

VWF von Willebrand factor

5 Types of device in contact with blood (as categorized in ISO 10993-1)

5.3 Implant devices

Implant devices are placed largely or entirely within the vascular system. Examples include but are not limited to

- annuloplasty rings,
- mechanical or tissue heart valves,
- prosthetic or tissue vascular grafts,
- circulatory support devices (ventricular-assist devices, artificial hearts, intra-aortic balloon pumps),
- inferior vena cava filters,
- embolization devices,
- endovascular grafts,
- implantable defibrillators and cardioverters,
- stents,
- arteriovenous shunts,
- blood monitors,

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- internal drug delivery catheters,
- pacemaker leads,
- intravascular membrane oxygenators (artificial lungs),
- leukocyte-removal filters.

6 Characterization of blood interactions**6.1 General requirements**

6.1.1 Figure 1 illustrates a decision tree that can be used to determine whether testing for interaction with blood is necessary.

Blood interactions can be classified into five categories based on the primary process or system being measured.

Tables 1 and 2 list examples of devices which contact circulating blood and the categories of testing appropriate to the device.

NOTE Since this is a horizontal International Standard, good rationales can be developed to justify the choice of category based on the device being characterized. Thrombosis testing is frequently the preferred method for device characterization. In many cases, rationales can be used to substitute some combination of coagulation, platelets, haematology and complement system testing for thrombosis testing.

For medical devices where a specific International Standard (vertical standard) exists, the biological evaluation requirements and test methods set forth in that vertical standard shall take precedence over the general requirements suggested in this part of ISO 10993.

6.1.2 Where possible, tests shall use an appropriate model or system which simulates the geometry and conditions of contact of the device with blood during clinical applications, including duration of contact, temperature, sterile condition and flow conditions. For devices of defined geometry, the ratio of test parameter (concentration per unit volume) to exposed surface area (cm²) shall be evaluated.

Only blood-contacting parts should be tested. The selected test methods and parameters should be in accordance with the current state of the art.

6.1.3 Controls shall be used unless their omission can be justified. Where possible, testing should include a relevant device already in clinical use or a well-characterized reference material [7].

Reference materials used should include negative and positive controls. All materials and devices tested shall meet all quality control and quality assurance specifications of the manufacturer and test laboratory. All materials and devices tested shall be identified as to source, manufacturer, grade and type.

6.1.4 Testing of materials which are candidates for components of a device may be conducted for screening purposes. However, such preliminary tests do not serve as a substitute for the requirement that the complete device or device component be tested under conditions which simulate or exaggerate clinical application.

6.1.5 Tests which do not simulate the conditions of a device during use may not predict accurately the nature of the blood/device interactions which can occur during clinical applications. For example, some short-term *in vitro* or *ex vivo* tests are poor predictors of long-term *in vivo* blood/device interactions [25], [26].

6.1.6 It follows from the above that devices whose intended use is *ex vivo* (external communication) should be tested *ex vivo* and devices whose intended use is *in vivo* (implants) should be tested *in vivo* in an animal model simulating as closely as possible conditions of clinical use.

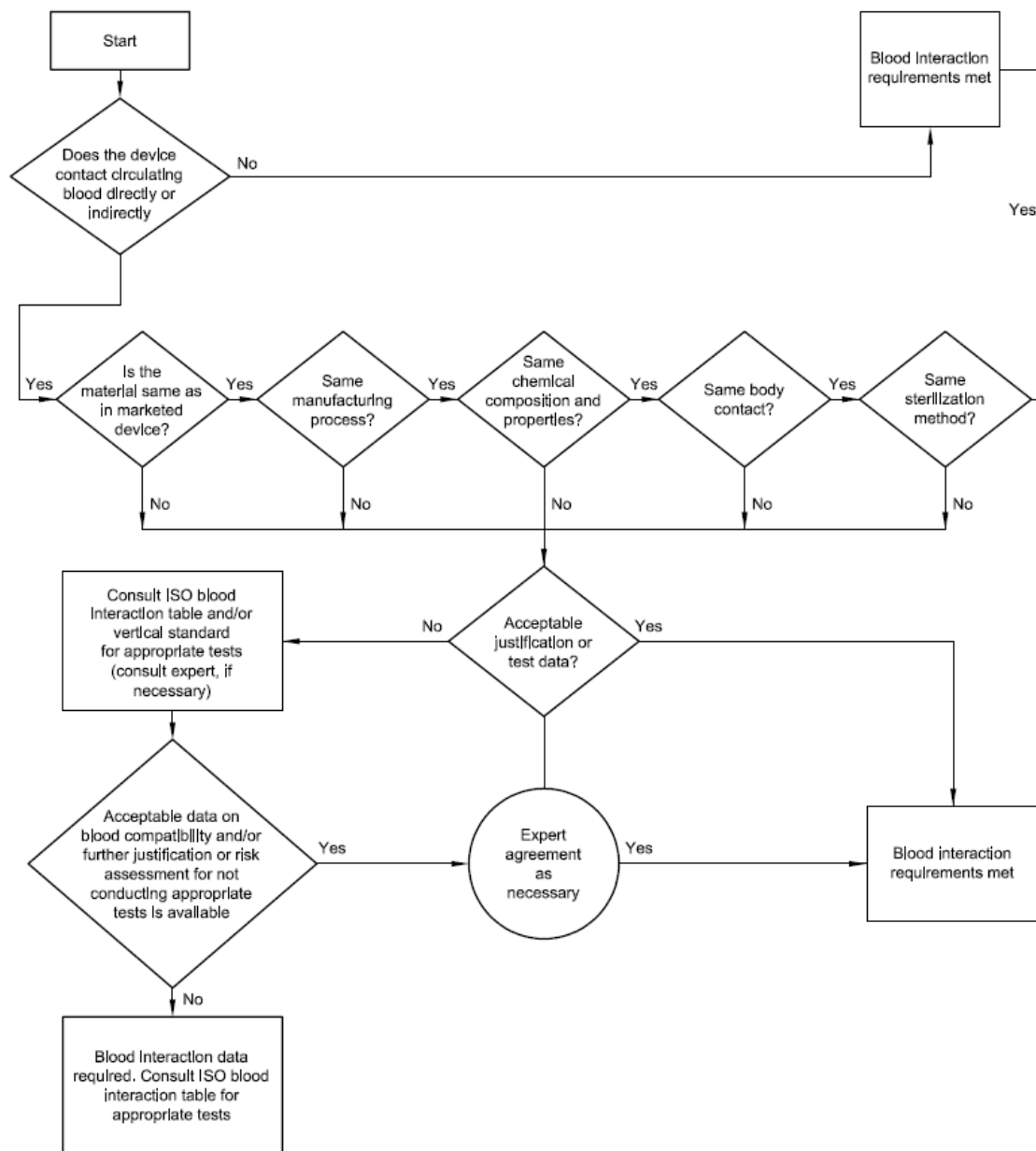


Figure 1 — Decision tree to determine whether testing for interaction with blood is necessary

Table 2 — Devices or device components which contact circulating blood and the categories of appropriate testing — Implant devices

Device examples	Test category				
	Thrombosis	Coagulation	Platelets	Haematology	Complement system
Annuloplasty rings, mechanical heart valves	x			x ^a	
Intra-aortic balloon pumps	x	x	x	x	x
Total artificial hearts, ventricular-assist devices	x			x	
Embolization devices				x ^a	
Endovascular grafts	x			x ^a	
Implantable defibrillators and cardioverters	x			x ^a	
Pacemaker leads	x			x ^a	
Leukocyte removal filter		x	x	x ^a	
Prosthetic (synthetic) vascular grafts and patches, including arteriovenous shunts	x			x ^a	
Stents	x			x ^a	
Tissue heart valves	x			x ^a	
Tissue vascular grafts and patches, including arteriovenous shunts	x			x ^a	
Vena cava filters	x			x ^a	
^a Haemolysis testing only.					

6.1.7 *In vitro* tests are regarded as useful in screening external communicating devices or implants, but may not be accurate predictors of blood/device interactions occurring upon prolonged or repeated exposure or permanent contact (see 6.3.1). Devices intended for non-contact use only do not require evaluation of blood/device interactions. Devices which come into very brief contact with circulating blood (e.g. lancets, hypodermic needles, capillary tubes) generally do not require blood/device interaction testing.

6.1.8 The recommendations in 6.1.6 and 6.1.7, together with clause 5, Figure 1 and Table 2, serve as a guide for the selection of tests listed in 6.2.1.

6.1.9 Disposable laboratory equipment used for the collection of blood and performance of *in vitro* tests on blood shall be evaluated to ascertain that there is no significant interference with the test being performed.

6.1.10 If tests are selected in the manner described and testing is conducted under conditions which simulate clinical applications, the results of such testing have the greatest probability of predicting clinical performance of devices. However, species differences and other factors may limit the predictability of any test.

6.1.11 Because of species differences in blood reactivity, human blood should be used where possible. When animal models are necessary, for example for evaluation of

devices used for prolonged or repeated exposure or permanent contact, species differences in blood reactivity shall be considered.

Blood values and reactivity in humans and non-human primates are very similar [26]. The use of animals such as the rabbit, pig, calf, sheep, or dog may also yield satisfactory results. Because species differences may be significant (for example platelet adhesion, thrombosis [20] and haemolysis tend to occur more readily in the canine species than in the human), all results of animal studies shall be interpreted with caution. The species used and the number of species used shall be justified (see ISO 10993-2).

NOTE The use of non-human primates for *in vivo* blood compatibility and medical device testing is prohibited by EU law

(86/609/EEC) and some national laws.

6.1.12 The use of anticoagulants in *in-vivo* and *ex-vivo* tests should be avoided unless the device is designed to perform in their presence. The choice and concentration of anticoagulant used influence blood/device interactions, and their selection shall be justified. Devices that are used with anticoagulants should be assessed using anticoagulants in the range of concentrations used clinically.

6.1.13 Modifications in a clinically accepted device shall be considered for their effect on blood/device interactions and clinical functions. Examples of such modifications include changes in design, geometry, changes in surface or bulk chemical composition of materials and changes in texture, porosity or other properties.

6.1.14 A sufficient number of replications of a test including suitable controls shall be performed to permit statistical evaluation of the data. The variability in some test methods requires that those tests be repeated a sufficient number of times to determine significance. In addition, repeated studies over an extended period of blood/device contact provide information about the time-dependence of the interactions.

6.2 Categories of tests and blood interactions

6.2.1 Recommended tests for interactions of devices with blood

Recommended tests are organized on the basis of the type of device according to Tables 3 and 4.

The tests are classified into the following five categories based on the primary process or system being measured:

- a) thrombosis (see 3.3);
- b) coagulation (see 3.4);
- c) platelets (see 3.5);

d) haematology (see 3.6);

e) complement system (see 3.7).

The principles and scientific bases for these tests are presented in annex B.

Table 4 — Test methods for implant devices

Test category	Method	Comments
Thrombosis	Scanning EM (platelet adhesion and aggregation); platelet and leukocyte morphology; fibrin	
	Percent occlusion	
	Flow reduction	
	Labelled antibodies to thrombotic components	
	Autopsy of devices (gross and microscopic); histopathology	
	Autopsy of distal organs (gross and microscopic); histopathology	
Coagulation	Specific coagulation factor assay; FPA, D-dimer, F ₁₊₂ , PAC-1, S-12, TAT	
	PTT(non-activated), PT, TT; plasma fibrinogen; FDP	
Platelets	PF-4, β -TG, thromboxane B ₂ ,	
	Platelet activation markers	
	Platelet microparticles	
	Gamma imaging of radiolabelled platelets; ¹¹¹ In labelled platelet survival	
	Platelet function analysis	
	Platelet count/adhesion	
	Platelet aggregation	
Haematology	Leukocyte count with or without differential;	
	Leukocyte activation	
	Haemolysis	
	Reticulocyte count; activation-specific release products of peripheral blood cells i.e. granulocytes)	
Complement system	C3a, C5a, TCC, Bb, iC3b, C4b, SC5b-9, CH 50, C3 convertase, C5 convertase	

6.2.4 Implant devices

After using Tables 1 and 2 to determine the relevant blood interaction category for a specific device type, Table 4 can be used as a guide to select the appropriate tests for implant devices as a function of the blood interactions appropriate for evaluation (see also 6.1.6). Test selection criteria depend on the specific device evaluated.

6.2.5 Indications and limitations

Immunoassays are available for human blood testing but are not generally available for other species. The human test kits usually do not cross-react with other species except for some non-human primates. Care should be taken when designing test systems to ensure that one is actually measuring activation due to the test material and not an artifact of the system. In vitro and ex vivo simulations with human blood often produce plasma levels of analytes that require a low, medium or high level of dilution, depending on experimental conditions, for measurement in the valid range of the immunoassay. Care should be taken to report only those results measured within the valid ranges of the assays. Care should also be taken to ensure that a range of dilutions of the sample tested is measured.

Discrepancies in evaluating blood/device interactions may occur because of inadequate materials characterization or inappropriate handling before blood tests are performed. For example, the studies may have relied on only one type of test or may have permitted the introduction of foreign material unrelated to the material or device under test.

Materials to be used in a low-flow (venous) environment may interact with blood quite differently when used in highflow (arterial) situations. Changes in design and/or flow conditions can alter the apparent in vivo haemocompatibility of a material.

6.3 Types of test

6.3.1 In vitro tests

Variables that shall be considered when using *in vitro* test methods include haematocrit, anticoagulants, sample collection, sample age, sample storage, aeration and pH, temperature, sequence of test versus control studies, surface-to-volume ratio, and fluid dynamic conditions (especially wall shear rate). Tests shall be performed with minimal delay, usually within 4 h, since some properties of blood change rapidly following collection.

6.3.3 In vivo tests

In vivo testing involves implanting the material or device in animals. Vascular patches, vascular grafts, prosthetic rings, heart valves and circulatory assist devices are examples of configurations used in *in vivo* testing. Patency (of a conduit) is the most common measure of success or failure for most *in vivo* experiments. The percent occlusion and thrombus mass are determined after the device is removed. The tendency of thrombi formed on a device to embolize to distal organs should be assessed by careful gross as well as microscopic examination of organs downstream from the device. In addition, histopathological evaluation of the surrounding tissue and organs is useful. The kidneys are especially prone to trap thrombi which have embolized

from devices implanted upstream from the renal arteries (for example ventricular-assist devices, artificial hearts, aortic prosthetic grafts) [19].

Methods to evaluate *in vivo* interactions without terminating the experiment are available. Arteriograms are used to determine graft patency or thrombus deposition on devices. Radioimaging can be used to monitor platelet deposition at various time periods *in vivo*; platelet survival and consumption can be used as indicators of blood/device interactions and passivation due to neointima formation or protein adsorption.

In some *in vivo* test systems, the material's properties may not be major determinants of the blood/device interactions. Rather, flow parameters, compliance, porosity and implant design may be more important than blood compatibility with the material itself. As an example, low flowrate systems may give substantially different results when compared to the same material evaluated in a high flowrate system. In such cases, test system performance *in vivo* should carry more importance than *in vitro* test results.

Annex A

(informative)

Preclinical evaluation of cardiovascular devices and prostheses

A.1 General considerations

A.1.1 Background

This annex provides background for selecting tests to evaluate the interactions of cardiovascular devices with blood. Clause 6 of this part of ISO 10993 contains guidance on when testing is necessary, what blood interaction categories might be appropriate for specific devices, and a list of tests for evaluating blood/device interactions of non-contact-, external communicating-, and implant-devices.

A.1.2 Classification

The following classification of blood/device interactions is provided as background.

a) Interactions which mainly affect the device and which may or may not have an undesirable effect on the subject are as follows:

- 1) adsorption of plasma proteins, lipids, calcium or other substances from the blood onto the surface of the device; or absorption of such substances into the device;
- 2) adhesion of platelets, leukocytes or erythrocytes onto the surface of the device, or absorption of their components into the device;
- 3) formation of pseudointima or tissue capsule on the surface of the device;
- 4) alterations in mechanical and other properties of the device.

b) Interactions which have a potentially undesirable effect on the animal or human are as follows:

- 1) activation of platelets, leukocytes or other cells, or activation of the coagulation, fibrinolytic, or complement pathways;
- 2) formation of thrombi on the device surface;
- 3) embolization of thrombotic or other material from the device's luminal surface to another site within the circulation;
- 4) injury to circulating blood cells resulting in anemia, haemolysis, leucopenia, thrombocytopenia or altered function of blood cells;
- 5) injury to cells and tissues adjacent to the device;
- 6) intimal hyperplasia or accumulation of other tissue on or adjacent to the device, resulting in reduced flow or affecting other functions of the device;
- 7) adhesion and growth of bacteria or other infectious agents on or near the device.

A.1.3 Advantages and limitations of animal and in vitro testing

Animal models permit continuous device monitoring and systematic controlled study of important variables.

However the choice of an animal model may be restricted by size requirements, the availability of certain species and cost. It is critical that the investigators be mindful of the physiological differences and similarities of the species chosen with those of the human, particularly those relating to coagulation, platelet functions and fibrinolysis, and the response to pharmacological agents such as anesthetics, anticoagulants, thrombolytic and antiplatelet agents, and antibiotics. Because of species differences in reactivity, subject differences in reactivity, and variable responses to different devices, data obtained from a single species should be interpreted with caution. Non-human primates such as baboons bear a close similarity to the human in haematological values, blood coagulation mechanism and cardiovascular system [30]. An additional advantage of a non-human primate is that many of the immunological probes for thrombosis developed for humans are suitable for use in other primates. These probes include PF-4, b-TG, FPA, TAT, and F1+2. The dog is a commonly used species and has provided useful information; however, device-related thrombosis in the dog tends to occur more readily than in the human, a difference which can be viewed as an advantage when evaluating this complication. The pig is generally regarded as a suitable animal model because of its haematological and cardiovascular similarities to the human. The effect of the surgical implant procedure on results should be kept in mind and appropriate controls included.

Because of species differences in haemostatic and haematological factors and activities, it is preferable to use human blood in in vitro tests whenever possible.

Thrombus formation is a dynamic process. Therefore in vitro testing is advisable to simulate as much as possible the dynamic conditions (for example shear forces of the blood/material interface) in which thrombosis occurs. Static tests may be useful in some cases for evaluating the interactions of blood with materials.

Since patients with cardiovascular devices may be receiving anticoagulant or antithrombotic drugs, it is important to simulate these conditions in vitro.

A.1.4 Test protocols for animal testing

Thrombosis, thromboembolism, bleeding and infection are the major deterrents to the use and further development of advanced cardiovascular prostheses. For devices with limited blood exposure (< 24 h), important measurements are related to the acute extent of variation of haematological, haemodynamic and performance variables, gross thrombus formation and possible embolism. With prolonged or repeated exposure, or permanent contact (> 24 h), emphasis is placed on serial measurement techniques that may yield information regarding the time course of thrombosis and thromboembolism, the consumption of circulating blood components, the development of intimal hyperplasia and infection. In both of the above exposure and contact categories, assessment of haemolysis and platelet function is important. Thrombus formation may be greatly influenced by surgical technique, variable time-dependent thrombolytic and embolic phenomena, superimposed device infections and possible alterations in exposed surfaces, for example intimal hyperplasia and endothelialization.

The consequences of the interaction of artificial surfaces with the blood can range from gross thrombosis and embolization to subtle effects such as accelerated consumption of haemostatic elements; the latter may be compensated (the total number of platelets consumed by the device is so small it does not affect the total platelet count) or lead to depletion of platelets or plasma coagulation factors (the device surface area is large enough to consume enough platelets for the total platelet count to be affected).

A.1.5 Test protocols for in vitro testing

In vitro testing allows for the performance of a sufficient number of tests for statistical evaluation without the sacrifice of animals and with relatively low costs. Measurements are related to the acute extent of variation of haematological, haemodynamic and performance variables, gross thrombus formation and complement activation.

The in vitro approach permits the study of the kinetics of various factors and activities, by varying the duration of exposure of material or devices to blood.

A.5 Ventricular-assist devices and total artificial hearts

These devices can induce considerable alteration in various blood components. Factors contributing to such effects include the large foreign surface area to which blood is exposed, the high flow regimes and the regions of disturbed flow such as turbulence or separated flow. Tests of such devices may include measurements of haemolysis, fibrinogen concentration, thrombin generation, platelet survival and activation, complement activation, and close monitoring of liver, renal, pulmonary and central nervous system functions. A detailed pathological examination at surgical retrieval is an important component of the evaluation [40], [41].

Annex B

(informative)

Laboratory tests — Principles, scientific basis and interpretation

B.1 General

B.1.1 Background

The principles and scientific basis of the tests listed in 6.2.1 are described here. Detailed methods are found in standard texts of laboratory medicine and clinical pathology. References [17] to [44], [46] to [49] and [59] describe tests which may be useful in the evaluation of blood/device interactions. Because of both biological variability and technical limitations, the accuracy of many of these tests is limited.

B.1.2 Principles for *in vitro* testing

Static and dynamic systems, e.g. circulatory loop and centrifugation systems, are used [50], [54].

B.1.3 Test conditions

In order for tests to be of use in the *in vitro* evaluation of blood/device interactions, anticoagulated blood or plasma collected from normal human subjects or experimental animals should first be exposed to the material or device under standardized conditions including time, temperature and flowrate. An aliquot of the exposed blood or plasma is then tested shortly after exposure. Conditions of exposure should be based on the intended use of the device. In the preparation of the test articles, it is essential to avoid activation or release of any component from blood before testing. However, the appropriate conditions depend on the device or material being tested and its intended end-use.

B.1.4 Classification

When evaluating externally communicating devices and implant devices while in their in-use position, blood is collected into an anticoagulant and the test is performed as described without a prior exposure stage. The tests are classified into five categories, as

defined in 6.2.1, according to the process or system being tested: thrombosis, coagulation, platelets and platelet functions, haematology and complement system.

B.2 Thrombosis

B.2.1 Percentage occlusion

Percentage occlusion is visually quantified after a device has been in use and has been removed. This is a measure of the severity of the thrombotic process in a conduit. Lack of occlusion does not necessarily eliminate the existence of a thrombotic process, since thrombi may have embolized or been dislodged before percentage occlusion is measured. Occlusion may be caused not only by thrombosis, but also by intimal hyperplasia, especially at perianastomotic sites in vascular grafts; microscopic examination is required to identify the nature of the occlusive process. Determinations of surface area covered by thrombus and thrombus-free surface area are semi-quantitative tests that can be used on a comparative basis.

B.2.2 Flow reduction

Flow (rate or volume) is measured after a period of use. Measurements may be performed either during use, or before and after use. Rationale and interpretation are the same as B.2.1.

B.2.3 Gravimetric analysis (thrombus mass)

This is conducted after removal of the device from the in-use position. Rationale and interpretation are as for B.2.1.

B.2.4 Light microscopy

By this technique, information can be obtained regarding the density of cells, cellular aggregates and fibrin adherent to materials, as well as the geographic distribution of these deposits on the materials or device. The method is semi-quantitative.

B.2.5 Pressure drop across device

This is measured before and after a period of use. Rationale and interpretation are as for B.2.1.

B.2.6 Scanning electron microscopy (SEM)

Rationale and interpretation are the same as B.2.4. This method has the advantage over B.2.4 of providing greater detail about fine structure of components being examined. Quantitative conclusions require sufficient replicate determinations to establish degree of reproducibility.

B.2.7 Antibody binding

Next to qualitative microscopic judgement of fibrin and platelet deposition on materials, a quantitative estimation is possible by measuring the amount of labelled antibody specific for fibrin(ogen) or platelet membrane receptors. For this purpose materials are washed after exposure to blood to delete non-adherent blood components prior to labelled antibody binding.

B.2.8 Retrieval and examination of device

This method is of great importance in evaluating the biological responses to implanted devices. The distribution, size and microscopic nature of cellular and proteinaceous deposits can best be determined by a careful and detailed examination. Proposed procedures have been published [40], [41].

B.2.9 Autopsy of distal organs

The rationale is to examine for distal effects of implanted devices. These effects include thromboembolism, and embolization of components of the device [43].

B.2.10 Imaging techniques — Angiography, intravascular ultrasound, Doppler ultrasound, CT

and MRI

Choices can be made among these methods to determine patency or degree of narrowing of a graft or other conduit and to detect thrombus deposition on devices during their *in vivo* performance.

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